PACE: Probabilistic Adaptive Contention Control for Non-Primary Channel Access in IEEE 802.11bn

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***Abstract*—Modern Wi-Fi deployments increasingly suffer from spectrum underutilization as overlapping Basic Service Sets (OBSSs) compete for primary channel access. IEEE 802.11bn introduces Non-Primary Channel Access (NPCA) to exploit these underutilized intervals, yet the standard contention mechanism is fundamentally mismatched to NPCA's constrained transmission windows: backoff counters routinely exceed the remaining opportunity duration, wasting slots that could otherwise carry data. We propose PACE, a probabilistic adaptive contention control scheme that enables stations to collectively converge toward a theoretically optimal transmission probability without central coordination. PACE combines a distributed multiplicative feedback rule—copying the successful sender's probability on solo receptions—with a prospective self-exclusion mechanism that silences stations whose frame length exceeds the remaining window. We evaluate PACE through simulation spanning a wide range of station counts and window lengths, comparing against Binary Exponential Backoff and several competing adaptive schemes under both homogeneous and heterogeneous frame-length distributions. PACE consistently outperforms standard backoff in channel utilization and achieves Pareto-dominance in the throughput–fairness tradeoff. The advantage is most pronounced when the available window is scarce relative to the contending population, precisely the regime in which standard backoff wastes the largest share of transmission opportunities. These results identify the tight-window regime as the critical operating condition for NPCA and establish PACE as a practical design reference for adaptive contention policy in next-generation Wi-Fi networks.**

***Index Terms*—Non-Primary Channel Access, IEEE 802.11bn, Probabilistic Access Control, OBSS, Adaptive Contention, Wi-Fi**

I. INTRODUCTION

Spectrum efficiency in next-generation Wi-Fi networks depends critically on how effectively stations exploit available channel time. In dense deployments, overlapping Basic Service Sets (OBSSs) routinely occupy the primary channel while adjacent channels remain idle, creating a persistent underutilization problem. IEEE 802.11bn addresses this gap by introducing Non-Primary Channel Access (NPCA), which allows stations to transmit opportunistically on non-primary channels during ongoing OBSS intervals [1], [2].

NPCA, however, presents a contention challenge distinct from conventional channel access. Each NPCA opportunity has a finite duration bounded by the ongoing OBSS transmission, and competing stations must coordinate their access within this constrained window. The Binary Exponential Backoff (BEB) mechanism was designed for open-ended contention: each collision doubles the backoff window, and the resulting delay frequently exceeds the remaining NPCA window duration. Affected stations are effectively silenced for the rest of the interval, leaving the channel idle even when other stations could have transmitted. In heterogeneous deployments where stations carry frames of varying lengths, the problem is further compounded: stations whose frame cannot fit in the remaining window continue to occupy contention slots despite having no viable transmission opportunity.

Consider a dense office environment where multiple stations detect an OBSS transmission and attempt to use the secondary channel concurrently. Early collisions trigger exponential backoff growth; within a few rounds, a large fraction of stations have accumulated backoff counts that exceed the remaining window duration. The channel spends the final slots idle, not because no station wanted to transmit, but because the contention mechanism pushed every backoff counter beyond the opportunity horizon.

Existing work on NPCA has focused on characterizing aggregate throughput and delay under the standard switching rule, taking Binary Exponential Backoff within the non-primary channel as given [2]–[4]. The contention dynamics within the finite NPCA window—and the mismatch between BEB's infinite-horizon design and the window's bounded duration—have not been addressed. Adaptive probabilistic schemes designed for infinite-horizon neighbor discovery, such as the ALOHA-like Neighbor Discovery (AND) algorithm [5], reduce overhead by avoiding explicit coordination, but their stationary-population assumption makes them ill-suited to a window that depletes over time. BEB's infinite-horizon design and AND's open-loop phase structure both fail to track the time-varying optimal access probability as the window shrinks and the set of viable stations changes (see Table I).

In this paper, we propose PACE, a Probabilistic Adaptive Contention control scheme for NPCA that enables stations to collectively converge toward the theoretically optimal transmission probability for each moment within the window, without central coordination or explicit inter-station signaling. PACE adapts the MIMD feedback rule of the Probabilistic Neighbor Discovery (PND) algorithm [6] to the finite-window NPCA context: stations copy the successful sender's probability on solo receptions, reduce on collision, and in-

TABLE I
COMPARISON OF CONTENTION CONTROL SCHEMES

Method	Mechanism	Objective	PPDU-aware	Signaling-free	NPCA target
AND [5]	Fixed-probability slotted ALOHA	Discover all neighbors in finite time without coordination	×	✓	×
PND [6]	Multiplicative feedback with solo-copy rule	Converge transmission probability to optimal without N	×	✓	×
Cali *et al.* [7]	p-persistent with contender estimation	Approach theoretical throughput limit in a distributed manner	×	✓	×
BLADE [8]	Hybrid increase / multiplicative decrease via access rate	Eliminate packet-delivery droughts for real-time Wi-Fi	×	✓	×
IYT [9]	Neighbor-ordered backoff interval assignment	Reduce collision probability and worst-case latency	×	✓	×
ACWB-DQN [10]	Deep Q-network contention window selection	Maximize throughput under varying network load	×	×	×
FC-MAPPO [11]	Fairness-constrained multi-agent policy optimization	Minimize collisions while ensuring inter-station fairness	×	×	×
PACE (proposed)	**Multiplicative feedback with PPDU-aware self-exclusion**	**Maximize channel utilization in finite NPCA window**	✓	✓	✓

crease on idle slots, causing distributed dynamics to track the throughput-optimal policy. PACE further adds a PPDU-aware self-exclusion rule that silences stations whose frame length exceeds the remaining window, eliminating a class of futile contention attempts that BEB would otherwise sustain.

The design of PACE is motivated by two requirements that are unique to NPCA's bounded-window structure: *(i)* the optimal transmission probability is time-varying and must track the depleting window rather than converge to a static equilibrium, and *(ii)* stations whose frame length exceeds the remaining window must proactively self-exclude to prevent futile contention that wastes viable slots. The main contributions of this paper are as follows:

- We derive the throughput-optimal transmission probability for finite-window NPCA, showing that it must continuously track the remaining window duration and the number of still-viable competing stations—a time-varying target that neither BEB nor existing infinite-horizon adaptive schemes can follow.
- We propose PACE, a fully distributed algorithm that addresses both requirements simultaneously: MIMD feedback drives the transmission probability toward the time-varying optimum *(i)*, while PPDU-aware self-exclusion eliminates futile contention by silencing stations that cannot complete a frame within the remaining window *(ii)*, all without inter-station signaling or knowledge of the total station count.
- Through extensive simulation under homogeneous and heterogeneous frame-length distributions, we demonstrate that PACE achieves significantly higher window utilization than BEB, is Pareto-dominant in the throughput–fairness tradeoff, and maintains proportional fairness between native and visiting stations.

The remainder of this paper is organized as follows. Section II reviews related work on NPCA and adaptive MAC protocols. Section III presents the system model. Section IV describes the PACE algorithm and its theoretical foundation. Section V presents simulation results, and Section VI concludes the paper.

II. RELATED WORK

Table I summarizes the key distinctions among existing contention control schemes and the proposed PACE.

A. Non-Primary Channel Access in IEEE 802.11bn

IEEE 802.11bn introduces NPCA as a core mechanism for exploiting idle secondary channels during OBSS transmissions [1], [12]. Wei *et al.* [2] extended Bianchi's DCF model to a multi-channel setting, establishing analytically that NPCA always outperforms the legacy baseline; a follow-up study [4] incorporated switching overhead and proposed a threshold policy to toggle between NPCA and legacy modes based on primary channel occupancy. Bellalta *et al.* [3] developed a Continuous-Time Markov Chain model for a multi-BSS topology, identifying a secondary-channel contention trade-off that can limit NPCA gains when the secondary channel is shared. Lee and Song [13] further showed that PPDU frame duration has a nonlinear effect on inter-BSS coexistence, and Song [14] applied deep reinforcement learning to the NPCA channel-activation decision.

Across these studies, however, the contention mechanism within the NPCA window is held fixed at Binary Exponential Backoff. How stations should adapt their access probability as the window depletes and viable stations drop out has not been studied.

B. Adaptive Contention Control in Wireless LANs

Bianchi [15] established that BEB operates below the theoretical throughput ceiling, and Cali *et al.* [7] showed that p-persistent access with dynamic contender estimation can approach this limit in a distributed manner. Subsequent work has diversified the adaptation mechanisms: BLADE [8] uses a HIMD policy driven by a universally observable channel signal; Wilhelmi *et al.* [9] impose structured transmission order via inter-BSS activity awareness; and learning-based approaches optimize the contention window through deep

Q-networks [10] or fairness-constrained multi-agent policy optimization [11].

Despite this breadth, all existing schemes assume stations contend over an open-ended horizon and target steady-state convergence. NPCA's bounded window demands a fundamentally different approach: the optimal transmission probability must evolve as the window shrinks and the viable station population changes. The MIMD feedback rule of PND [6] is the closest mechanism in spirit, but was designed for infinite-horizon neighbor discovery and provides no accounting for bounded window duration or frame-length viability.

III. System Model

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IV. Pace Algorithm

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V. Performance Evaluation

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VI. Conclusion

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